

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	A	A	B	D		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	A,C,D	A,B,C	A,C	A,B,C	B,C	A,B,C,D
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	0.00	7.41	0.08	4.87	3.50	3.00
SECTION-II (ii)	Q.	7	8	9			
	A.	200	50	4			

PART-2 : CHEMISTRY

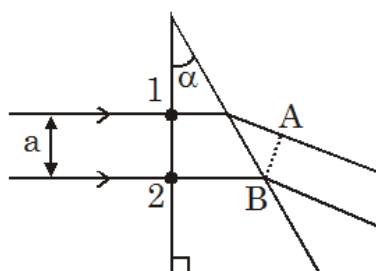
SECTION-I (i)	Q.	1	2	3	4		
	A.	A	D	C	A		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	B,C	B,C,D	A,B,D	A,B,C,D	A,B,C,D	A,B,C
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	7.00	10.00	6.00	4.00	46.00	2.00
SECTION-II (ii)	Q.	7	8	9			
	A.	3	4	8			

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	C	C	B	A		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	A,B	B,C	A,D	A,B,D	A,C,D	A,B,C
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	3.00	1.50	120.00	9.00	3.50	0.00
SECTION-II (ii)	Q.	7	8	9			
	A.	9	20	13			

HINT – SHEET**PART-1 : PHYSICS****SECTION-I (i)**

1. Ans (A)



$$0(1A) = 0(2B)$$

$$\therefore \Delta P = 0$$

$$\Rightarrow \Delta \phi = 0$$

2. Ans (A)

The intensity of light reflected from upper surface is

$$I_1 = I_0 \times 25\% = I_0 \times \frac{25}{100} = \frac{I_0}{4}$$

The intensity of transmitted light from upper surface is

$$I = I_0 - \frac{I_0}{4} = \frac{3I_0}{4}$$

$\therefore$ The intensity of reflected light from lower surface is

$$I_2 = \frac{3I_0}{4} \times \frac{50}{100} = \frac{3I_0}{8}$$

$$\therefore \frac{I_{\max}}{I_{\min}} = \frac{(\sqrt{I_1} + \sqrt{I_2})^2}{(\sqrt{I_1} - \sqrt{I_2})^2}$$

$$\therefore \frac{I_{\max}}{I_{\min}} = \frac{\left(\sqrt{\frac{I_0}{4}} + \sqrt{\frac{3I_0}{8}}\right)^2}{\left(\sqrt{\frac{I_0}{4}} - \sqrt{\frac{3I_0}{8}}\right)^2} = \frac{\left(\frac{1}{2} + \sqrt{\frac{3}{8}}\right)^2}{\left(\frac{1}{2} - \sqrt{\frac{3}{8}}\right)^2}$$

3. **Ans (B)**

$$P = \frac{h}{\lambda} \Rightarrow P_{\text{net}} = \frac{Nh}{\lambda}$$

$$\Rightarrow F = \frac{Nh}{\Delta t \lambda} = \frac{nh}{\lambda} = ma$$

4. **Ans (D)**

Theoretical

PART-1 : PHYSICS

SECTION-I (ii)

5. **Ans (A,C,D)**

Theoretical

6. **Ans (A,B,C)**

$$(A) \lambda_{\min} = \frac{hc}{eE}$$

$$(B) E_K - E_L = \frac{hc}{e\lambda_{K\alpha}}$$

$$\text{where } E_K = 25.51 \text{ \& } E_L = 3.56$$

$$(C) E_K - E_M = \frac{hc}{e\lambda_{K\beta}}$$

7. **Ans (A,C)**

$$T_y = T_0 \frac{dy}{dx}$$

8. **Ans (A,B,C)**

The minimum energy required to remove the electron from K-level is, 3×10^{-15} J let V is the potential difference required, then

$$eV = 3 \times 10^{-15} \text{ J}$$

$$\Rightarrow V = \frac{3 \times 10^{-15}}{1.6 \times 10^{-19}} \text{ V} = 18750 \text{ V olt}$$

The energy released due to the transition of electron from L-level to K-level is,

$$\Delta E = 3 \times 10^{-15} - 2 \times 10^{-16} = 28 \times 10^{-16} \text{ J}$$

The wavelength of em waves is,

$$\lambda = \frac{hc}{\Delta E} = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{28 \times 10^{-16}} \text{ m}$$

$$= 7.104 \times 10^{-11} \text{ m}$$

The minimum energy required to eject the electron from M-level is 3×10^{-17} J

The energy used in ejecting the electron from M-level is, 28×10^{-16} J.

So, the KE of emitted electrons from M-level is,

$$K = (28 \times 10^{-16} - 3 \times 10^{-17}) \text{ J}$$

$$K = 277 \times 10^{-17} \text{ J.}$$

9. **Ans (B,C)**

$$(B) \frac{dM}{M} = \frac{0.003}{0.3} \times 100\% = 1\%$$

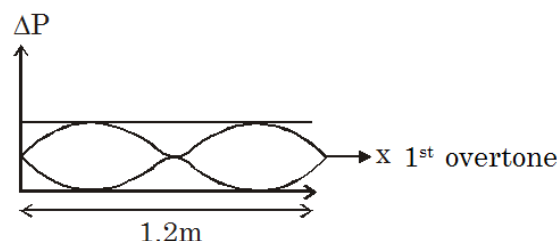
$$(C) I = \frac{1}{2} M(a^2 + b^2)$$

$$\frac{dI}{I} = \frac{dM}{M} + \frac{2(ada + bdb)}{a^2 + b^2}$$

$$= 1\% + \frac{2(0.04 + 0.02)}{4^2 + 2^2} \times 150\% = 1.6\%$$

10. **Ans (A,B,C,D)**

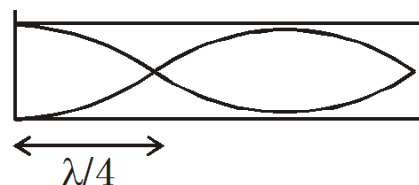
(A) & (B)



$$\therefore S_N = 0.3, 0.9$$

$$P_N = 0, 0.6, 1.2$$

(C & D)



$$\frac{3\lambda}{4} = 1.2$$

$$\therefore \frac{\lambda}{4} = 0.4$$

PART-1 : PHYSICS

SECTION-II (i)

1. **Ans (0.00)**

$$\Delta P = \mu_1 S_1 S_2 + \mu_2 t + \mu_3 (S_2 Q - t) - \mu_1 S_1 S_3 - \mu_3 S_3 Q$$

$$O = \mu_1 (S_1 S_2 - S_1 S_3) - (\mu_3 - \mu_1) t + \mu_3 (S_2 Q - S_3 Q)$$

$$O = \mu_1 d \tan Q - (\mu_3 - \mu_2) t + \mu_3 (S_2 Q - S_3 Q)$$

$$\& \tan \theta = \frac{d}{D} \therefore S_2 Q - S_3 Q = 0$$

$\therefore$ Central maxima is at P

2. **Ans (7.41)**

$$\Delta P = \mu_1 (S_1 S_2 - S_1 S_3) + \mu_3 (S_2 Q - S_3 Q)$$

$$O = \mu_1 \frac{d^2}{D} - \mu_3 d \frac{x}{D}$$

3. **Ans (0.08)**

$$\rho \quad [p - <p>]$$

$$1.8 \quad 0.1$$

$$2.0 \quad 0.1$$

$$2.0 \quad 0.1$$

$$1.9 \quad 0$$

$$1.8 \quad 0.1$$

$$<p> = \frac{9.5}{5} = 1.9 \quad \frac{0.4}{5} = 0.08$$

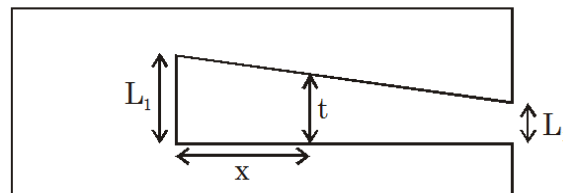
4. **Ans (4.87)**

$$V = \frac{m}{\rho} \Rightarrow \frac{dV}{V} = \frac{dm}{m} + \frac{d\rho}{\rho}$$

$$= \left(\frac{1}{152} + \frac{0.08}{1.9} \right) 100\% = 4.87$$

5. **Ans (3.5)**

From the theory mentioned in passage at the ends of cavity fringes will form, and as number of dark fringes is greater than number of bright fringes so ends will be the location of dark fringes. Thickness of the cavity at a distance x from the left end would be



$$t = L_2 + \frac{L_1 - L_2}{L} \times x$$

For left end, $2\mu \times L_1 = n\lambda_0$ [For dark fringe]

For right end, $2\mu L_2 = (n - 7)\lambda_0$

$$\Rightarrow L_1 - L_2 = \frac{7\lambda_0}{2\mu}$$

6. **Ans (3.00)**

For 4th dark fringe from left end

$$2\mu t = (n - 3)\lambda_0$$

$$\Rightarrow 2\mu \left[L_2 + \frac{L_1 - L_2}{L} x \right] = 2\mu L_1 - 3\lambda_0$$

$$\Rightarrow x = \frac{4L}{7}$$

PART-1 : PHYSICS

SECTION-II (ii)

7. **Ans (200)**

$$0 = \frac{P}{\lambda} - mv$$

$$\therefore E = \frac{1}{2}mv^2$$

8. **Ans (50)**

20 division of vernier scale = 39 mm

$$1 \text{ division of vernier scale} = \frac{39}{20} = 1.95 \text{ mm}$$

LC of main scale = 1 mm

$$\therefore \text{LC of instrument} = 2 - 1.95 = 0.05 \text{ mm}$$

9. Ans (4)

$$L = \frac{\lambda_1}{2} = 5 \dots\dots(i)$$

$$C = f\lambda_1 = 2(10) \dots\dots(ii)$$

$$\frac{\lambda_2}{4} = 1.25 \dots\dots(iii)$$

$$C = f\lambda_2$$

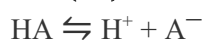
$$20 = f5$$

$$\therefore f = 4$$

PART-2 : CHEMISTRY

SECTION-I (i)

3. Ans (C)



$$\alpha = \sqrt{\frac{K_a}{C}} \Rightarrow K_a = 4 \times 10^{-8}$$

$$\Rightarrow H^+ = \sqrt{K_a C}, pH = 5 - \log 2 = 4.7$$

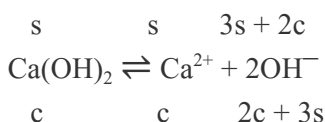
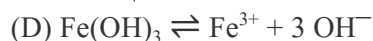
PART-2 : CHEMISTRY

SECTION-I (ii)

7. Ans (A,B,D)

$$(A) [H^+] = \sqrt{K_a C} = \sqrt{10^{-4} \times 0.1} = 10^{-2.5},$$

$$[OH^-] = \sqrt{\frac{K_w}{K_a} C} = 10^{-5.5}$$

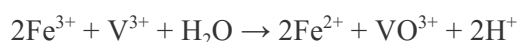


$$K_{sp} = s(3s + 2c)^3$$

$$s = \frac{K_{sp}}{8c^3} = \frac{4 \times 10^{-38}}{8 \times (10^{-2})^3}$$

8. Ans (A,B,C,D)

Cell reaction



$$E_{cell} = 0.43 - \frac{0.06}{2} \log \frac{[Fe^{+2}]^2 [VO^{+3}] [H^+]^2}{[Fe^{+3}]^2 [V^{+3}]}$$

$$= 0.52 \text{ volt}$$

PART-2 : CHEMISTRY

SECTION-II (i)

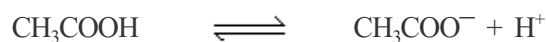
1. Ans (B)

In original solution

$$[H^+] = \sqrt{K_a \times C} = \sqrt{10^{-6} \times 1} = 10^{-3} \text{ M}$$

2. Ans (A)

Let resulting solution (After dilution) have concentration Co



$$pH = 2 \times 3 = 6 \quad [H^+] = x = 10^{-6}$$

$$K_a = \frac{[CH_3COO^-][H^+]}{[CH_3COOH]} = \frac{x \cdot x}{(Co - x)}$$

$$10^{-6} = \frac{(10^{-6})^2}{Co - 10^{-6}}$$

$$Co = 2 \times 10^{-6} \text{ M}$$

Now, for CH_3COOH

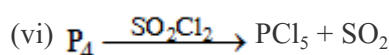
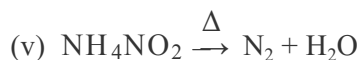
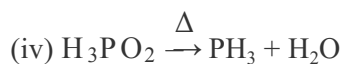
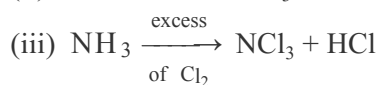
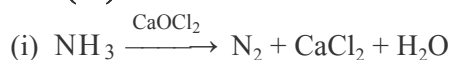
$$1 \text{ M} \times 1 \text{ litre} = 2 \times 10^{-6} \text{ M} \times V$$

$$V = 5 \times 10^5 \text{ litre.}$$

PART-2 : CHEMISTRY

SECTION-II (ii)

7. Ans (3)



9. Ans (8)

2 millimoles causes coagulation of 250 ml

$\therefore$ for coagulation of 1000 ml $Fe(OH)_3$,
millimoles of $Fe(OH)_3$ required is 8.

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (C)

Using cross product of three vectors

$$\begin{aligned} \hat{i} \times [(\vec{a} - \hat{j}) \times \hat{i}] \\ \Rightarrow (\hat{i} \cdot \hat{i})(\vec{a} - \hat{j}) - (\hat{i} \cdot (\vec{a} - \hat{j}))\hat{i} \\ \Rightarrow \vec{a} - \hat{j} - (\vec{a} \cdot \hat{i})\hat{i} \end{aligned}$$

$$\text{Let } \vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$$

$$\Rightarrow \vec{a} - \hat{j} - (a_1\hat{i}) \quad \dots(1)$$

similarly :

$$\begin{aligned} \hat{j} \times [(\vec{a} - \hat{k}) \times \hat{j}] \\ \Rightarrow \vec{a} - \hat{k} - a_2\hat{j} \quad \dots(2) \end{aligned}$$

$$\text{And : } \hat{k} \times [(\vec{a} - \hat{i}) \times \hat{k}]$$

$$\vec{a} - \hat{i} - a_3\hat{k} \quad \dots(3)$$

Adding (1) (2) and (3)

$$3\vec{a} - (\hat{i} + \hat{j} + \hat{k}) - (a_1\hat{i} + a_2\hat{j} + a_3\hat{k}) = 0$$

$$3\vec{a} - (\hat{i} + \hat{j} + \hat{k}) - \vec{a} = 0$$

$$\vec{a} = \frac{\hat{i} + \hat{j} + \hat{k}}{2}$$

$$\therefore \vec{a} \cdot \vec{b} = \left(\frac{\hat{i} + \hat{j} + \hat{k}}{2} \right) \cdot 2(\hat{i} + \hat{j} + \hat{k}) = 3$$

2. Ans (C)

$$2|z|^2 + |z|^2 + 3|z|^2 - |z|^2 = 6|z|$$

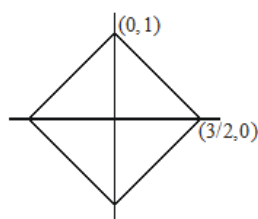
$$2|z(z + \bar{z})| + 3|z(z - \bar{z})| = 6|z|$$

$$2|z||z + \bar{z}| + 3|z||z - \bar{z}| = 6|z|$$

$$2|z + \bar{z}| + 3|z - \bar{z}| = 6$$

$$4|x| + 6|y| = 6$$

$$\frac{|x|}{\frac{3}{2}} + \frac{|y|}{2} = 1$$



$$\text{Area} = 4 \times \frac{1}{2} \times \frac{3}{2} \times 1$$

$$\Rightarrow 3 \text{ unit}^2$$

3. Ans (B)

The total number of coins in the box is $N + 7$.
The total value of the selected coins will be greater than or equal to one rupee and fifty paise if the selected coins are

- 1 fifty paise coin, 4 twenty-five paise coins
- 2 fifty paise coins, 3 twenty-five paise coins
- 2 fifty paise coins, 2 twenty-five coins and 1 coin out of the N five paise and ten paise coins.

If E = total value of the 5 elected coins is greater than or equal to one rupee and fifty paise, then

$$n(E) = {}^2C_1 \cdot {}^5C_4 \cdot {}^NC_0 + {}^2C_2 \cdot {}^5C_3 \cdot {}^NC_0 + {}^2C_2 \cdot {}^5C_2 \cdot {}^NC_1$$

Hence, required probability

$$= 1 - P(E) = 1 - \frac{n(E)}{n(S)} = 1 - \frac{10(N+2)}{N + {}^7C_5}$$

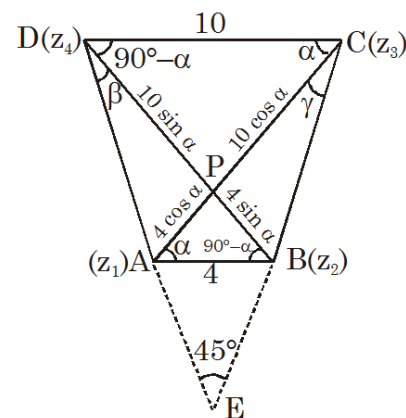
4. Ans (A)

$$\frac{{}^{20}C_5 \times {}^4C_2}{20 \times 19 \times 18 \times 17 \times 16} = \frac{1}{20}$$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. Ans (A,B)



We have $\angle C + \angle D + \angle E = 180^\circ$

$$\therefore \beta + \gamma = 45^\circ$$

From the figure we can say that

$$\tan \beta = \frac{2}{5} \cot \alpha, \quad \tan \gamma = \frac{2}{5} \tan \alpha$$

Since $\beta + \gamma = 45^\circ$

$$\therefore (1 + \tan \beta) (1 + \tan \gamma) = 2$$

$$\left(1 + \frac{2}{5} \cot \alpha\right) \left(1 + \frac{2}{5} \tan \alpha\right) = 2$$

$$\therefore \sin \alpha \cos \alpha = \frac{10}{21}$$

$$\therefore \sin 2\alpha = \frac{20}{21}$$

Area of trapezium is the sum of area of 4 triangle.

$$= \frac{1}{2} (16 \sin \alpha \cos \alpha + 40 \sin \alpha \cos \alpha + 100 \sin \alpha \cos \alpha + 40 \sin \alpha \cos \alpha)$$

$$= \frac{196}{2} \sin \alpha \cos \alpha = \frac{196}{2} \times \frac{10}{21} = \frac{140}{3}$$

$$|CP - DP| = |10 (\cos \alpha - \sin \alpha)| = \frac{10}{\sqrt{21}}$$

$$\text{In } (\Delta PCB) = \frac{1}{2} 40 \sin \alpha \cos \alpha = \frac{200}{21}$$

6. **Ans (B,C)**

$$(Z_1 - Z_0) (\bar{Z}_1 - \bar{Z}_0) = 16R^2$$

$$\Rightarrow Z_1 \bar{Z}_1 - Z_0 \bar{Z}_1 - \bar{Z}_0 Z_1 + |Z_0|^2 = 16R^2$$

...(1)

$$\left(\frac{1}{\bar{Z}_1} - Z_0\right) \left(\frac{1}{Z_1} - \bar{Z}_0\right) = 256R^2$$

$$\frac{1}{Z_1 \bar{Z}_1} - \frac{Z_0}{Z_1} - \frac{\bar{Z}_0}{\bar{Z}_1} + |Z_0|^2 = 256R^2$$

$$\text{put } |Z_1|^2 = k$$

$$1 - Z_0 \bar{Z}_1 - \bar{Z}_0 Z_1 + K |Z_0|^2 = 256R^2 K$$

$$(1) - (2)$$

$$\Rightarrow (K - 1) + (1 - K) |Z_0|^2 = 16R^2 (1 - 16K)$$

$$\Rightarrow (1 - K) + (|Z_0|^2 - 1) = 16R^2 (1 - 16K)$$

$$\Rightarrow (1 - K) = (1 - 16K)$$

$$\Rightarrow K = \frac{31}{511}$$

7. **Ans (A,D)**

$$|a| = |b| = 1$$

$$a \cdot b = 0; \text{ (as } a \perp b)$$

$$c = \alpha a + \beta b + \gamma(a \times b) \quad \dots(i)$$

Taking dot product by a,

$$a \cdot c = \alpha |a|^2 + \beta(a \cdot b) + \gamma[a \cdot (a \times b)]$$

$$\Rightarrow |a| \cdot |c| \cos \theta = \alpha \cdot 1 + 0 + 0$$

$$\Rightarrow 1 \cdot |1| \cdot \cos \theta = \alpha$$

Taking dot product of (i) by b

$$b \cdot c = b \cdot a + \beta |b|^2 + \gamma[b \cdot (a \times b)]$$

$$\Rightarrow |b| \cdot |c| \cos \theta = 0 + \beta \cdot 1 + 0$$

$$\therefore \beta = 1.1. \cos \theta = \cos \theta$$

$$|c|^2 = 1 \Rightarrow \alpha^2 + \beta^2 + \gamma^2 = 1$$

$$\Rightarrow \cos^2 \theta + \cos^2 \theta + \gamma^2 = 1$$

$$\therefore \gamma^2 = 1 - 2\cos^2 \theta = -\cos 2\theta$$

$$\text{Hence, } \alpha = \beta = \cos \theta, \gamma^2 = -\cos 2\theta$$

8. **Ans (A,B,D)**

$$\text{Let } P(A \cup B) = x$$

$$\Rightarrow P(A) = \frac{2x}{3}, P(B) = \frac{x}{2} \text{ \&}$$

$$P(A \cap B) = \frac{2x}{6} \cdot P(B) = \frac{x^2}{6}$$

$$\Rightarrow x = \frac{2x}{3} + \frac{x}{2} - \frac{x^2}{6}$$

$$\Rightarrow x = P(A \cup B) = 1 \text{ \& } P(A \cap B) = \frac{1}{6}$$

9. **Ans (A,C,D)**

$$P_k = \frac{2}{3} P_{k-1} + \frac{1}{3} (1 - P_{k-1}) = \frac{1}{3} (1 + P_{k-1})$$

$$\Rightarrow 3P_k = 1 + P_{k-1}$$

$$P_1 = \frac{2}{3}; P_2 = \frac{5}{9}; P_3 = \frac{14}{27}; P_4 = \frac{41}{81};$$

$$P_5 = \frac{122}{243}$$

10. Ans (A,B,C)

$$(\bar{x} \times \bar{y}) \times \bar{b} = \bar{a} \times \bar{b}$$

$$\Rightarrow (\bar{x} \cdot \bar{b}) \cdot \bar{y} - (\bar{y} \cdot \bar{b}) \bar{x} = \bar{a} \times \bar{b}$$

$$\Rightarrow \gamma \cdot \bar{y} - (\bar{y} \cdot \bar{b}) \cdot \bar{x} = \bar{a} \times \bar{b} \quad (\because \bar{x} \cdot \bar{b} = \gamma)$$

$$\bar{y} \cdot \bar{b} = \bar{y} \cdot (\bar{y} \times \bar{z}) = 0$$

$$\Rightarrow \bar{y} = \frac{\bar{a} \times \bar{b}}{\gamma}$$

$$\Rightarrow (\bar{x} \times \bar{y}) \times \bar{y} = \bar{a} \times \bar{y}$$

$$\Rightarrow (\bar{x} \cdot \bar{y}) \cdot \bar{y} - (\bar{y} \cdot \bar{y}) \bar{x} = \bar{a} \times \bar{y}$$

$$\Rightarrow \bar{y} - |\bar{y}|^2 \bar{x} = \bar{a} \times \bar{y}$$

$$(\because \bar{x} \cdot \bar{y} = 1)$$

$$\Rightarrow \bar{x} = \frac{1}{y^2} [\bar{y} - \bar{a} \times \bar{y}]$$

$$(\bar{y} \times \bar{z}) \times \bar{y} = \bar{b} \times \bar{y}$$

$$\Rightarrow y^2 \bar{z} - (\bar{z} \cdot \bar{y}) \cdot \bar{y} = \bar{b} \times \bar{y}$$

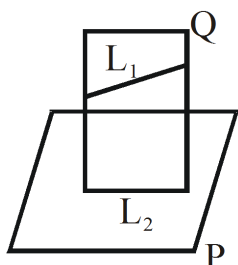
$$\Rightarrow y^2 \bar{z} - \bar{y} = \bar{b} \times \bar{y}$$

$$\Rightarrow \bar{z} = \frac{1}{y^2} [\bar{y} + \bar{b} \times \bar{y}]$$

PART-3 : MATHEMATICS

SECTION-II (i)

1. Ans (3.00)



$$\text{Let } L_1 : \frac{x}{2} = \frac{y-1}{2} = \frac{z-1}{1}$$

and projection of L_1 on

$$\text{plane P is } L_2 : \frac{x}{1} = \frac{y-1}{1} = \frac{z-1}{-1}$$

vector normal to plane Q containing line L_1 and L_2 is

$$\vec{n}_1 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 2 & 1 \\ 1 & 1 & -1 \end{vmatrix} = -3\hat{i} + 3\hat{j}$$

Now direction Ratios of normal to plane 'P' is

$$\vec{n}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -3 & 3 & 0 \\ 1 & 1 & -1 \end{vmatrix} = -3\hat{i} - \hat{j} - 6\hat{k}$$

Hence equation of plane P is

$$-3(x-0) - 3(y-1) - 6(z-1) = 0$$

$$-3x - 3y + 3 - 6z + 6 = 0$$

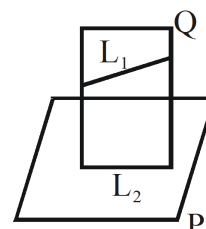
$$3x - 3y - 6z + 9 = 0$$

$$x + y + 2z - 3 = 0$$

Distance of plane 'P' from origin

$$\left| \frac{0+0+0-3}{\sqrt{6}} \right| = \frac{3}{\sqrt{6}} = \sqrt{\frac{3}{2}}$$

2. Ans (1.50)



$$\text{Let } L_1 : \frac{x}{2} = \frac{y-1}{2} = \frac{z-1}{1}$$

and projection of L_1 on

$$\text{plane P is } L_2 : \frac{x}{1} = \frac{y-1}{1} = \frac{z-1}{-1}$$

vector normal to plane Q containing line L_1 and L_2 is

$$\vec{n}_1 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 2 & 1 \\ 1 & 1 & -1 \end{vmatrix} = -3\hat{i} + 3\hat{j}$$

Now direction Ratios of normal to plane 'P' is

$$\vec{n}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -3 & 3 & 0 \\ 1 & 1 & -1 \end{vmatrix} = -3\hat{i} - \hat{j} - 6\hat{k}$$

Hence equation of plane P is

$$-3(x-0) - 3(y-1) - 6(z-1) = 0$$

$$-3x - 3y + 3 - 6z + 6 = 0$$

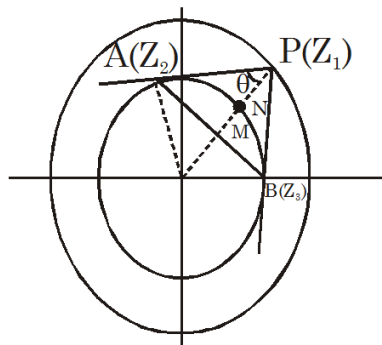
$$3x - 3y - 6z + 9 = 0$$

$$x + y + 2z - 3 = 0$$

Distance of plane 'P' from origin

$$\left| \frac{0+0+0-3}{\sqrt{6}} \right| = \frac{3}{\sqrt{6}} = \sqrt{\frac{3}{2}}$$

3. Ans (120.00)



$$OA = OB = 1; OP = 2$$

$$\sin \angle OPA = \frac{1}{2}, \quad \angle OPA = \frac{\pi}{6}, \quad \angle APB = \frac{\pi}{3}$$

OM $\perp$ AB so $\triangle APB$ is equilateral.

$$OM = \frac{1}{2}, \quad MN = \frac{1}{2}, \quad PN = 1 \text{ so}$$

centroid on $|z| = 1$ (A)

$$\left| \frac{z_1 + z_2 + z_3}{3} \right| = 1$$

$$|z_1 + z_2 + z_3|^2 = 9$$

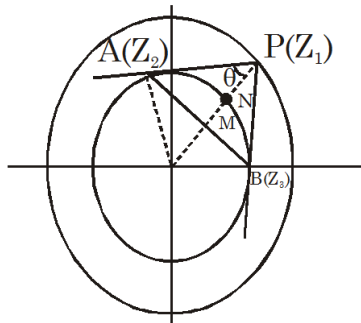
$$(z_1 + z_2 + z_3)(\bar{z}_1 + \bar{z}_2 + \bar{z}_3) = 9$$

$$z_1 \bar{z}_1 = 4 \quad z_2 \bar{z}_2 = 1 \quad z_3 \bar{z}_3 = 1$$

$$\text{or } \left(\frac{4}{z_1} + \frac{1}{z_2} + \frac{1}{z_3} \right) \left(\frac{4}{\bar{z}_1} + \frac{1}{\bar{z}_2} + \frac{1}{\bar{z}_3} \right) = 9 \quad \dots(C)$$

$$\angle AOB = \pi - \frac{\pi}{3} = \frac{2\pi}{3} \quad \dots(B)$$

4. Ans (9.00)



$$OA = OB = 1; OP = 2$$

$$\sin \angle OPA = \frac{1}{2}, \quad \angle OPA = \frac{\pi}{6}, \quad \angle APB = \frac{\pi}{3}$$

OM $\perp$ AB so $\triangle APB$ is equilateral.

$$OM = \frac{1}{2}, \quad MN = \frac{1}{2}, \quad PN = 1 \text{ so}$$

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$$(z_1 + z_2 + z_3)(\bar{z}_1 + \bar{z}_2 + \bar{z}_3) = 9$$

$$z_1 \bar{z}_1 = 4 \quad z_2 \bar{z}_2 = 1 \quad z_3 \bar{z}_3 = 1$$

$$\text{or } \left(\frac{4}{z_1} + \frac{1}{z_2} + \frac{1}{z_3} \right) \left(\frac{4}{\bar{z}_1} + \frac{1}{\bar{z}_2} + \frac{1}{\bar{z}_3} \right) = 9 \quad \dots(C)$$

$$\angle AOB = \pi - \frac{\pi}{3} = \frac{2\pi}{3} \quad \dots(B)$$

5. Ans (3.50)

$$\left(\frac{z+1}{z} \right)^8 = \left| \frac{z+1}{z} \right|$$

$$\frac{z+1}{z} = t$$

$$t^8 = |t|$$

taking modulus

$$|t|^8 = |t| \Rightarrow |t| = 0 \text{ or } 1$$

but given question $|t| \neq 0$

$$|t| = 1$$

$$t^8 = 1$$

$$\left(\frac{z+1}{z} \right)^8 = 1$$

$$\frac{z+1}{z} = (1)^{\frac{1}{8}}$$

$$\frac{z+1}{z} = \cos \frac{2k\pi}{8} + i \sin \frac{2k\pi}{8}$$

$$k = 0, 1, 2, 3, \dots, 7 \quad \text{let } \frac{2k\pi}{8} = \theta$$

$$\frac{z+1}{z} = \cos \theta + i \sin \theta$$

$$z+1 = (\cos \theta + i \sin \theta)z$$

$$z = \frac{1}{\cos \theta + i \sin \theta - 1}$$

$$= \frac{(\cos \theta - 1 - i \sin \theta)}{((\cos \theta - 1) + i \sin \theta)((\cos \theta - 1) - i \sin \theta)}$$

$$z = \frac{\cos \theta - 1 - i \sin \theta}{(\cos \theta - 1)^2 + \sin^2 \theta}$$

$$z = \frac{\cos \theta - 1 - i \sin \theta}{\cos^2 \theta + 1 - 2 \cos \theta + \sin^2 \theta}$$

$$z = \frac{(\cos \theta - 1) - i \sin \theta}{2 - 2 \cos \theta}$$

$$z = -\frac{1}{2} - \frac{i \sin \theta}{2(1 - \cos \theta)}$$

$$\Rightarrow z = -\frac{1}{2} - \frac{i}{2} \frac{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}{\frac{2 \sin^2 \theta}{2}}$$

$$z = -\frac{1}{2} - \frac{i}{2} \cot \frac{\theta}{2} \quad \text{where } \theta = \frac{2K\pi}{8}$$

$$z = -\frac{1}{2} - \frac{i}{2} \cot \frac{K\pi}{8}$$

All z will lie on $\operatorname{Re}(z) = -\frac{1}{2}$, which is a line

So all are collinear.

$k=0$ is not possible

$$z_1 = -\frac{1}{2} - \frac{i}{2} \cot \frac{\pi}{8}$$

$$z_2 = -\frac{1}{2} - \frac{i}{2} \cot \frac{2\pi}{8}$$

$$z_7 = -\frac{1}{2} - \frac{i}{2} \cot \frac{7\pi}{8}$$

$$\text{so, } \sum_{r=1}^n \operatorname{Re}(z_r) = -\frac{7}{2}$$

$$\sum_{r=1}^n \operatorname{Im}(z_r) = 0$$

6. **Ans (0.00)**

$$\left(\frac{z+1}{z}\right)^8 = \left|\frac{z+1}{z}\right|$$

$$\frac{z+1}{z} = t$$

$$t^8 = |t|$$

taking modulus

$$|t|^8 = |t| \Rightarrow |t| = 0 \text{ or } 1$$

but given question $|t| \neq 0$

$$|t| = 1$$

$$t^8 = 1$$

$$\left(\frac{z+1}{z}\right)^8 = 1$$

$$\frac{z+1}{z} = (1)^{\frac{1}{8}}$$

$$\frac{z+1}{z} = \cos \frac{2k\pi}{8} + i \sin \frac{2k\pi}{8}$$

$$k = 0, 1, 2, 3, \dots, 7 \quad \text{let } \frac{2k\pi}{8} = \theta$$

$$\frac{z+1}{z} = \cos \theta + i \sin \theta$$

$$z+1 = (\cos \theta + i \sin \theta)z$$

$$z = \frac{1}{\cos \theta + i \sin \theta - 1}$$

$$= \frac{(\cos \theta - 1 - i \sin \theta)}{((\cos \theta - 1) + i \sin \theta)((\cos \theta - 1) - i \sin \theta)}$$

$$z = \frac{\cos \theta - 1 - i \sin \theta}{(\cos \theta - 1)^2 + \sin^2 \theta}$$

$$z = \frac{\cos \theta - 1 - i \sin \theta}{\cos^2 \theta + 1 - 2 \cos \theta + \sin^2 \theta}$$

$$z = \frac{(\cos \theta - 1) - i \sin \theta}{2 - 2 \cos \theta}$$

$$z = -\frac{1}{2} - \frac{i \sin \theta}{2(1 - \cos \theta)}$$

$$\Rightarrow z = -\frac{1}{2} - \frac{i}{2} \frac{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}{\frac{2 \sin^2 \theta}{2}}$$

$$z = -\frac{1}{2} - \frac{i}{2} \cot \frac{\theta}{2} \quad \text{where } \theta = \frac{2K\pi}{8}$$

$$z = -\frac{1}{2} - \frac{i}{2} \cot \frac{K\pi}{8}$$

All z will lie on $\operatorname{Re}(z) = -\frac{1}{2}$, which is a line

So all are collinear.

$k=0$ is not possible

$$z_1 = -\frac{1}{2} - \frac{i}{2} \cot \frac{\pi}{8}$$

$$z_2 = -\frac{1}{2} - \frac{i}{2} \cot \frac{2\pi}{8}$$

$$z_7 = -\frac{1}{2} - \frac{i}{2} \cot \frac{7\pi}{8}$$

$$\text{so, } \sum_{r=1}^n \operatorname{Re}(z_r) = -\frac{7}{2}$$

$$\sum_{r=1}^n \operatorname{Im}(z_r) = 0$$

PART-3 : MATHEMATICS

SECTION-II (ii)

7. **Ans (9)**

$$h(x) = \frac{1}{(x-\omega)(x-\omega^2)} = \frac{1}{(\omega-\omega^2)} \left(\frac{1}{x-\omega} - \frac{1}{x-\omega^2} \right)$$

$$h^{(36)}(x) = \frac{1}{(\omega-\omega^2)} \left[\frac{(-1)^{36} \times 36!}{(x-\omega)^{37}} - \frac{(-1)^{36} (36!)}{(x-\omega^2)^{37}} \right]$$

$$= \frac{1}{\omega-\omega^2} (36!) \left(\frac{-1}{\omega} + \frac{1}{\omega^2} \right) = 36!$$

$$36 = 2^2 3^2$$

so, number of divisors = 9.

8. Ans (20)

$$P\left(\frac{W}{L}\right) = \frac{2}{3}, P\left(\frac{L}{W}\right) = \frac{1}{3}$$

$$P\left(\frac{W}{L}\right) = \frac{1}{3}, P\left(\frac{L}{L}\right) = \frac{2}{3}$$

Required probability

$$= P(WWWL) + P(WWLW) + P(WLWW) + P(LWWW)$$

$$= \left(\frac{2}{3}\right)^3 \times \frac{1}{3} + \left(\frac{2}{3}\right)^2 \times \left(\frac{1}{3}\right)^2 + \left(\frac{2}{3}\right) \left(\frac{1}{3}\right)^2 \times \left(\frac{2}{3}\right)$$

$$+ \left(\frac{1}{3}\right)^2 \times \left(\frac{2}{3}\right)^2$$

$$= \frac{8 + 4 + 4 + 4}{3^4} = \frac{20}{81}$$

9. Ans (13)

 Let E_1 = Event that A wrote a plus sign.

 E_2 = Event that A wrote a minus sign.

 E = Event that the reference observes a plus sign.

$$\text{Given } P(E_1) = \frac{1}{3} \Rightarrow P(E_2) = \frac{2}{3}$$

 $P(E/E_1)$ = Probability that none of B, C, D change sign + Probability that exactly two of B, C, D change sign.

$$= \frac{1}{27} + 3 \left(\frac{1}{3} \times \frac{2}{3} \times \frac{2}{3} \right) = \frac{13}{27}$$

 $P(E/E_2)$ = Probability that all of B, C, D change the sign + Probability that exactly one of them change the sign.

$$= \frac{8}{27} + 3 \times \left(\frac{2}{3} \times \frac{1}{3} \times \frac{1}{3} \right) = \frac{14}{27}$$

$$\therefore P(E_1/E) = \frac{13}{41} \text{ Using Baye's theorem.}$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Main)

UNIT TEST

24-12-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TEAS, TTAS, TOAS, TNAS, TRAS, TMAS, TIAS, I(A) & LIVE-I(A)]

ANSWER KEY**PAPER-2****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	D	A	C	C	A	B	C	B	C
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	A	C	A	A	D	C	C	B	C
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	3.10 to 3.16	5.00	1.00	3.00	3.00	0.77 to 0.78	5.00	5.00	4.00 to 5.00	4.00

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	D	D	D	A	B	D	D	C	D
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	C	B	D	D	A	C	C	D	C	A
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	10.00	0.00	8.00	2.50	4.00	4.00	3.00	3.00	3.00	7.00

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	B	C	B	D	C	B	C	B	A
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	B	B	B	D	C	C	B	B	C
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	7.00	6.00	8.00	2.00	7.00	3.00	1.00	11.00	4.00	3.50

HINT – SHEET**PART-1 : PHYSICS****SECTION-I**4. **Ans (C)**

$$\tan 57^\circ = \frac{\mu_g}{\mu_a}$$

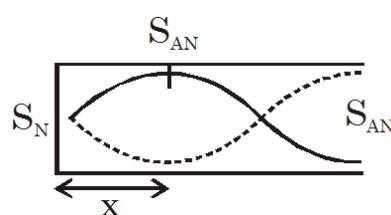
$$\tan \theta = \frac{\mu_g}{\mu_w}$$

$$\theta < 57^\circ$$

5. **Ans (C)**

$$\frac{V_{\text{gas}}}{V_{\text{air}}} = \frac{f\lambda_{\text{gas}}}{f\lambda_{\text{air}}} = \frac{\Delta \ell'}{\Delta \ell}$$

$$V_{\text{gas}} = \frac{1000}{3} \times \frac{3}{2} \frac{\Delta \ell}{\Delta \ell} = 500 \text{ ms}^{-1}$$

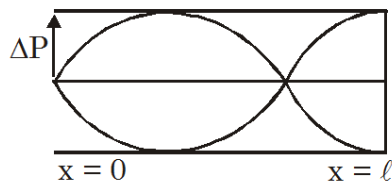
6. **Ans (A)**

$$\frac{3\lambda}{4} = 1.2$$

$$\lambda = 1.6 \text{ m}$$

$$x = \frac{\lambda}{4} = 0.4 \text{ m}$$

7. **Ans (B)**



$$f = \frac{3v}{4\ell}$$

$$f = \frac{3v}{4\ell} = \frac{3 \times 330}{4 \times 550}$$

$$\ell = \frac{9}{20} \text{ m} = 45 \text{ cm}$$

The amplitude of pressure variation in the pipe is $a = |\Delta P_0 \sin kx|$

8. **Ans (C)**

$$\frac{dN}{dt} = \alpha - \lambda N$$

$$\int_{N_0}^N \frac{dN}{\alpha - \lambda N} = \int_0^t dt$$

$$\frac{\alpha - \lambda N}{\alpha - \lambda N_0} = -\lambda t$$

$$\alpha - \lambda N = (\alpha - \lambda N_0) e^{-\lambda t}$$

$$N = \frac{1}{\lambda} [\alpha - (\alpha - \lambda N_0) e^{-\lambda t}]$$

$$\alpha = 2N_0\lambda$$

$$N = \frac{1}{\lambda} \left[2N_0\lambda - \frac{N_0\lambda}{2} \right] = \frac{3}{2} N_0$$

9. **Ans (B)**

$$I \propto \frac{1}{r^2}$$

If r is doubled no. is reduced to $\left(\frac{1}{4}\right)^{\text{th}}$

10. **Ans (C)**

$$\text{Reading} = 2 \times 1 \text{ mm} + (67 - 4) \times 0.01$$

$$= 2 \text{ mm} + 0.63 \text{ mm} = 2.63 \text{ mm}$$

11. **Ans (A)**

Minima will be formed at O, if

$$SS_1 + S_1O - SS_2 - S_2O = \frac{n\lambda}{2}, n = 1, 2, \dots$$

For minimum value of d , $n = 1$

$$\therefore \frac{\lambda}{4} = \sqrt{1 + d^2} - 1$$

$$(1 + d^2)^{1/2} - 1 = \frac{\lambda}{4}$$

$$1 + \frac{d^2}{2} - 1 = \frac{\lambda}{4} \text{ (Negative smaller terms)}$$

$$\Rightarrow d = \sqrt{\frac{\lambda}{2}}; \beta = \frac{D\lambda}{d} = \sqrt{2\lambda}$$

12. **Ans (A)**

$$E_{PQR} = \text{Constant}$$

$$E_{PQ} = \text{Constant}$$

$$P = T \frac{\partial y}{\partial t} \frac{\partial y}{\partial x}$$

$$\text{at Q, } \frac{\partial y}{\partial x} = 0$$

$$P = 0$$

13. **Ans (C)**

If n = number of moles

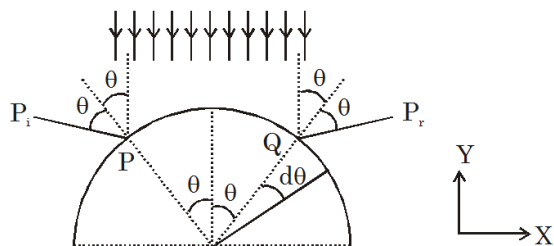
$$\text{Rate of decay} = \lambda nN = \frac{nN \ln 2}{T}$$

$$\text{Rate of release of energy} = E \left[\frac{dN}{dt} \right]$$

$$p = \eta E \left[\frac{dN}{dt} \right] = \eta \frac{nNE \ln 2}{T}$$

14. Ans (A)

Let us consider the cross-section of the cylinder as shown in figure.



Consider two identical small area elements, located symmetrically at P & Q as shown in figure

The momentum of the reflected light photons along the X-axis is zero and it is only along Y-axis.

Number of incident photons per unit normal area per unit time on the cylinder is. $n_1 = \frac{I\lambda}{hc}$.

The momentum imparted by the incident photons per unit area per unit time is,

$$\Delta P_i = n_1 \times \frac{h}{\lambda} = \frac{I}{c}.$$

The momentum imparted by the incident photons per unit time is,

$$P_i = \int_{-\pi/2}^{\pi/2} \frac{I}{c} \times (h \times R d\theta) \times \cos \theta = \frac{2IhR}{c}$$

[along -ve Y-axis]

The momentum imparted by the reflected photon per unit time is,

$$P_r = \int_{-\pi/2}^{\pi/2} r \times n_1 \times \frac{h}{c} \times \cos 2\theta \times (hR d\theta) \times \cos \theta$$

[along +ve Y-axis]

$$= \frac{2}{3} \times \frac{rIhR}{c}$$

where r is reflection coefficient of surface.

Momentum transferred per unit time to surface,

$$\frac{\Delta P}{\Delta t} = P_f - P_i = \left(\frac{2r}{3} + 2 \right) \frac{IhR}{c}$$

For part a, $r = 1$

For part b, $r = 0.8$

15. Ans (A)

$$f = \frac{1}{2\ell} \sqrt{\frac{T}{m}};$$

$$\text{In air : } T = mg = \rho Vg$$

$$\therefore f = \frac{1}{2\ell} \sqrt{\frac{\rho Vg}{m}}$$

$$\text{In water : } T = mg - \text{upthrust}$$

$$= V\rho g - \frac{V}{2}\rho, \quad g = \frac{Vg}{2}(2\rho - \rho)$$

$$\therefore f' = \frac{1}{2\ell} \sqrt{\frac{\frac{Vg}{2}(2\rho - \rho)}{m}}$$

$$= \frac{1}{2\ell} \sqrt{\frac{Vg\rho}{m}} \sqrt{\frac{(2\rho - \rho)}{2\rho}}$$

$$\frac{f'}{f} = \sqrt{\frac{2\rho - \rho}{2\rho}}$$

$$f' = f \left(\frac{2\rho - \rho}{2\rho} \right)^{1/2}$$

$$300 \left(\frac{2\rho - 1}{2\rho} \right)^{1/2} \text{ Hz}$$

18. Ans (C)

$$r \propto h^2$$

$$v \propto \frac{1}{h}$$

$$E \propto \frac{1}{h^2}$$

$$mvr = \frac{nh}{2\pi}$$

19. Ans (B)

$$r \propto \frac{1}{qm}$$

20. Ans (C)

$$E \propto q^2m$$

$$27E_0 - \frac{27E_0}{9} = \frac{hc}{\lambda}$$

$$27E_0 \times \frac{8}{9} = PC$$

$$P = \frac{24E_0}{C}$$

PART-1 : PHYSICS

SECTION-II

1. **Ans (3.10 to 3.16)**

$$\Delta m = 80.9166 + 0.00054858 - 80.9163$$

$$Q = (\Delta m)931 = E = 0.79 \text{ MeV}$$

$$P = \frac{E}{C}$$

$$mv = P = \frac{E}{C}$$

$$v = \frac{E}{mc} = \frac{0.79 \times 10^6 \times 1.6 \times 10^{-19}}{80.9163 \times 1.67 \times 10^{-27} \times 3 \times 10^8}$$

$$v = 3117 \text{ m/s}$$

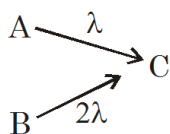
2. **Ans (5)**

$$b_A = \left(\frac{340 - 10}{340 - 10 + 10} \right) 85$$

$$b_B = \left(\frac{340 + 10}{340 + 10 - 10} \right) 85$$

$$b_{\text{beat}} = |b_A - b_B| = 5$$

3. **Ans (1)**



$$N_A = N_0 e^{-\lambda t} = \frac{N_0}{2}$$

$$N_B = N_0 e^{-2\lambda t} = \frac{N_0}{4}$$

$$\frac{dN_c}{dt} = \lambda_A N_A + \lambda_B N_B$$

$$= \frac{N_0 \lambda}{2} + \frac{N_0}{4} 2\lambda = \frac{N_0 \lambda}{2} + \frac{N_0 \lambda}{2} = N_0 \lambda$$

4. **Ans (3.00)**

From Wien's displacement law and Stefan Boltzman law the wavelength of the peak radiation is halved.

$$\text{So, } \lambda = 3000 \text{ \AA}$$

$$\frac{hc}{\lambda} - \frac{\alpha}{100} = 13.6 \left[\frac{1}{4} - \frac{1}{16} \right]$$

$$\frac{hc}{\lambda} - \frac{\alpha}{100} = 2.55$$

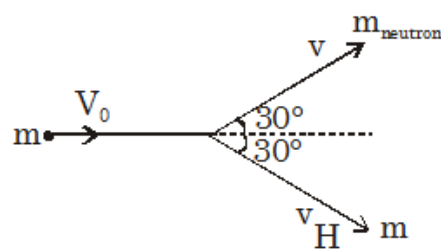
$$\frac{12420}{3000} - \frac{\alpha}{100} = 2.55$$

$$\frac{\alpha}{100} = 4.14 - 2.55 = 1.59$$

$$\alpha = 159 = 53x$$

$$x = 3$$

5. **Ans (3)**



$$mv_0 = mv \cos 30^\circ \times 2$$

$$mv_0 = mv \sqrt{3}$$

$$v = \frac{v_0}{\sqrt{3}}$$

$$10.2 \text{ eV} = \frac{1}{2} mv_0^2 - \left(\frac{1}{2} mv^2 \times 2 \right)$$

$$= \frac{1}{2} k - mv^2 = k - \frac{mv_0^2}{3}$$

$$= k - \frac{2k}{3} = \frac{k}{3}$$

$$k = 3 \times 10.2 \text{ eV}$$

6. **Ans (0.77 to 0.78)**

From conservation of energy

$$m_n c^2 = m_p c^2 + k_p + m_e c^2 + k_e + m_v c^2 + k$$

$$939.5656 = 938.2723 + 0.5109 + 0.0004 + (k_p + k_e)$$

$$[\because m_v c^2 + k_v = 0.0004 \text{ MeV}]$$

$$\Rightarrow k_p + k_e = 0.0004 \text{ MeV}$$

$$\Rightarrow \frac{p^2}{2m_p} + \frac{p^2}{2m_e} = 0.7820 \text{ MeV}$$

$$\Rightarrow \frac{p^2}{2m_e} \left[1 + \frac{m_e}{m_p} \right] = 0.7820$$

$$k_e = \left(\frac{m_p}{m_p + m_e} \right) \times 0.7820 \simeq 0.7820 \text{ MeV}$$

7. **Ans (5.00)**

$$x = A^3 - AC$$

$$dx = 3A^2 dA - CdA - AdC$$

$$= (3A^2 - C)dA - AdC$$

$$\% \text{ error} = \frac{dx}{x} \times 100 = \frac{(3A^2 - C)dA + AdC}{A^3 - AC} \times 100$$

$$= \frac{(3(20)^2 - 300)(0.1) + 20(2)}{(20)^3 - (20)(300)} \times 100 = 6.5$$

8. Ans (5.00)

$$\rho_B v_g = R_1 = 4.00 \pm 0.02$$

$$\rho_B v_g - \rho_w v_g = R_2 = 1.00 \pm 0.02$$

$$\frac{\rho_B v_g - \rho_w v_g}{\rho_B v_g} = \frac{R_2}{R_1}$$

$$1 - \frac{\rho_w}{\rho_B} = \frac{R_2}{R_1} \Rightarrow \frac{\rho_w}{\rho_B} = 1 - \frac{R_2}{R_1}$$

$$\frac{1}{y} = \frac{\rho_B}{\rho_w} = \frac{R_1}{R_1 - R_2}$$

$$\Rightarrow y = \frac{R_1 - R_2}{R_1} = 1 - \frac{R_2}{R_1}$$

$$dy = \frac{-R_1 dR_2 + R_2 dR_1}{R_1^2}$$

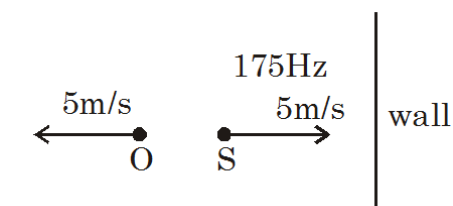
$$\% \text{ error} = \frac{dy}{y} \times 100$$

$$= \left(\frac{R_2 dR_1 + R_1 dR_2}{R_1^2} \right) \left(\frac{R_1}{R_1 - R_2} \right) \times 100$$

$$= \frac{2.5}{3} = \frac{\rho}{6}$$

$$\rho = 5$$

9. Ans (4 to 5)



$$f_1 = \left(\frac{355 - 5}{355 + 5} \right) 175$$

$$f_2 = \left(\frac{355 - 5}{355 - 5} \right) \times 175 \times \left(\frac{355 - 5}{355} \right)$$

$$f_{\text{beat}} = |f_1 - f_2| = 4.86$$

10. Ans (4.00)

$$\beta = \frac{i_C}{i_B} = 80$$

$$V_{R_1} = V_{cc} - V_{BE} = 6 - 1 = 5$$

$$i_B = \frac{V_{R_1}}{R_1} = \frac{5}{100k\Omega}$$

$$i_C = 80 \times \frac{5}{100 \times 1000}$$

$$V_{R_2} = i_C R_2 = 80 \times \frac{5}{100 \times 1000} \times 1000$$

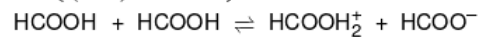
$$V_{R_2} = 4$$

PART-2 : CHEMISTRY

SECTION-I

8. Ans (D)

$$[\text{HCOOH}] = \frac{1 \text{ mol}}{\left\{ \left(\frac{46}{1.15} \right) \times 10^{-3} \ell \right\}} = 25M$$



$$\begin{array}{ccccccc} t = 0 & 25M & 25M & 0 & 0 \\ \text{At equili.} & 25 - 10^{-3} & 10^{-3} & 10^{-3} & 10^{-3} \end{array}$$

$$\alpha (\%) = \frac{10^{-3}}{25} \times 100 = 0.004\%$$

9. Ans (C)

$$K_h = \frac{(1.6 \times 10^{-4})^2}{0.01} = 2.56 \times 10^{-6}$$

$$K_h = \frac{K_w}{K_b} \Rightarrow K_b = 3.9 \times 10^{-9}$$

10. Ans (D)

Refer theory

11. Ans (C)

Factual

13. Ans (D)

(FeSO₄·(NH₄)₂SO₄·6H₂O) Mohr salt

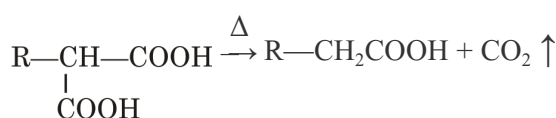
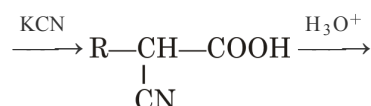
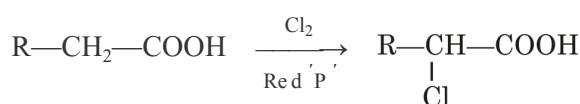
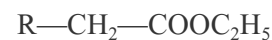
$$\lambda_m^\circ(\text{salt}) = \lambda_m^\circ(\text{Fe}^{2+}) + 2\lambda_m^\circ(\text{NH}_4^+)$$

$$+ 2\lambda_m^\circ(\text{SO}_4^{2-}) = a + 2b + 2c$$

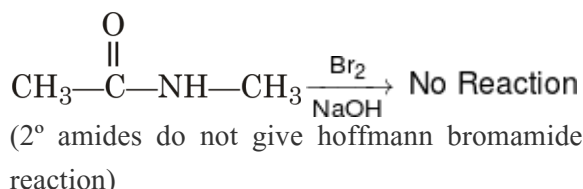
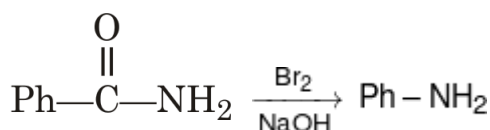
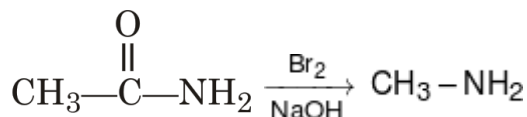
$$\lambda_{\text{eq}}^\circ(\text{salt}) = \frac{\lambda_m^\circ(\text{salt})}{\nu f} = \frac{a + 2b + 2c}{4}$$

16. Ans (C)

Acids on reaction with alcohols give sweet smelling esters.



17. Ans (C)



PART-2 : CHEMISTRY

SECTION-II

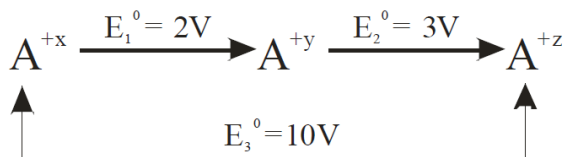
2. Ans (0.00)

$x = 3$ (K, Rb, Cs)

$y = 2$ (Mg, Be)

$u = 5$ (Concentrated solution of all alkali metals in liquid ammonia is blue coloured solution)

3. Ans (8.00)



$$\text{A}^{+x} + (x - y) \rightarrow \text{A}^{+y} \quad 2012$$

$$\text{A}^{+y} + (y - z) \rightarrow \text{A}^{+z} \quad 3V \quad 10$$

$$-(x - z) \times 10 = (x - y) \times 2 - (y - z) \times 3$$

$$(x - z) 10 = \frac{y}{x} + 7 \left(\frac{z}{x} \right)$$

$$10x - 10z = 2x - 2y + 3y - 3z$$

$$8x - 7z = y$$

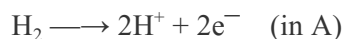
$$8 - 7 \left(\frac{z}{x} \right) = \left(\frac{y}{x} \right)$$

$$8 = 7 \frac{z}{x} + \frac{y}{x}$$

4. Ans (2.50)

25 mg is the gold no. [gold no. is mg]

5. Ans (4.00)



$$E_1 = 0.06 \text{ pH} \Rightarrow \text{pH} = \frac{E_1}{0.06}$$

$$\text{pH} = \text{pKa} + \log \frac{x}{y}$$

$$\frac{E_1}{0.06} = \text{pKa} + \log \frac{x}{y} \quad (1)$$

$$\text{pH} = \frac{E_2}{0.06} = (\text{in B})$$

$$\frac{E_2}{0.06} = \text{pKa} + \log \frac{y}{x} \quad (2)$$

PART-3 : MATHEMATICS

SECTION-I

1. Ans (D)

$v_1 = \text{variance of } \{13, 16, 19, \dots, 103\}$

$= \text{variance of } \{3, 6, 9, \dots, 93\}$

$= 9 (\text{variance of } \{1, 2, 3, \dots, 31\})$

$v_2 = \text{variance of } \{20, 26, 32, \dots, 200\}$

$= \text{variance of } \{6, 12, 18, \dots, 186\}$

$= 36 (\text{variance of } \{1, 2, 3, \dots, 31\})$

$$\therefore \frac{v_1}{v_2} = \frac{1}{4}$$

2. Ans (B)

p	$\sim p$	q	$p \rightarrow q$	$q \vee \sim p$	$(p \rightarrow q) \longleftrightarrow (q \vee \sim p)$
T	F	T	T	T	T
T	F	F	F	F	T
F	T	T	T	T	T
F	T	F	T	T	T

3. Ans (C)

Number of ways to choose A and B = 2^n . $2^n = 4^n$.

The number of subsets which contain exactly r elements in nC_r . $\therefore$ Number of ways to choose A and B such that they have same number of elements is

$$\left({}^nC_0 \right)^2 + \left({}^nC_1 \right)^2 + \left({}^nC_2 \right)^2 + \dots + \left({}^nC_n \right)^2 = {}^{2n}C_n$$

$$\therefore \text{Required } \frac{{}^{2n}C_n}{2^{2n}} = \frac{1.3.5 \dots (2n-1)}{2^n \cdot n!}$$

4. Ans (B)

cases

- (i) 2 steps forward, 3 steps backward
- (ii) 3 steps forward, 2 steps backward
- (iii) 2 steps 'as it is', 2 steps forward, 1 step back.
- (iv) 2 steps 'as it is', 2 steps back, 1 step forward.
- (v) 4 steps 'as it is', 1 step back.
- (vi) 4 steps 'as it is', 1 step forward.

5. Ans (D)

$|z - 3| + |z + 1| = 8$ represent an ellipse with centre $z = 1 + 0i$ and focus $3 + 0i, -1 + 0i$
 $|z - 3| + |z + 1| \geq 8$ represent ellipse and outer part of ellipse

$$\Rightarrow |z - 1|_{\min} = \frac{\text{length of minor axis}}{2} = 2\sqrt{3}$$

6. Ans (C)

Let $z = x + iy$

$$-8((x + iy)^2 + (x - iy)^2) + 20(x^2 + y^2) = 144$$

$$-8(2(x^2 - y^2)) + 20(x^2 + y^2) = 144$$

$$4x^2 + 36y^2 = 144$$

$$\frac{x^2}{36} + \frac{y^2}{4} = 1$$

$$a = 6, b = 2$$

$$\text{Area} = \pi ab = 12\pi$$

7. Ans (B)

$$(5\vec{a} - 3\vec{b}) \cdot (3\vec{a} + 4\vec{b}) = 0$$

$$15 + 11\vec{a} \cdot \vec{b} - 12 = 0$$

$$\vec{a} \cdot \vec{b} = \frac{-3}{11}$$

$$\cos \theta = \frac{-3}{11} \Rightarrow \sin \theta = \frac{\sqrt{112}}{11}$$

$$|\vec{a} \times \vec{b}| = \frac{\sqrt{112}}{11}$$

8. Ans (C)

$$\vec{b} \times (\vec{v} \times \vec{b}) = \vec{b} \times \vec{c}$$

$$4\vec{v} - (\vec{b} \cdot \vec{v})\vec{b} = \vec{b} \times \vec{c}$$

$$|\vec{v} \times \vec{b}| = |\vec{c}|$$

$$\Rightarrow |\vec{v}| |\vec{b}| \sin \theta = |\vec{c}|$$

$$2 |\vec{v}| \sin \theta = \sqrt{3}$$

$$\sin \theta = \frac{\sqrt{3}}{2}$$

$$\cos \theta = \frac{1}{2}$$

$$\Rightarrow 4\vec{v} - \left(2 \times 1 \times \frac{1}{2}\right)\vec{b} = \vec{b} \times \vec{c}$$

$$\vec{v} = \frac{1}{4}(\vec{b} + \vec{b} \times \vec{c})$$

9. Ans (B)

Let equation of plane is $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$,

where $A \equiv (a, 0, 0)$, $B \equiv (0, b, 0)$, $C \equiv (0, 0, c)$

plane passes through $(1, -2, 3)$

$$\therefore \frac{1}{a} - \frac{2}{b} + \frac{3}{c} = 1$$

hence locus of point (a, b, c) is

$$yz - 2xz + 3xy = xyz$$

10. Ans (A)

By Baye's Theorem

$$P\left(\frac{\text{scooter}}{\text{claims}}\right) = \frac{\frac{50}{150} \times .02}{\frac{50}{150} \times .02 + \frac{40}{150} \times .06 + \frac{60}{150} \times .15} = \frac{5}{62}$$

11. Ans (B)

$$T_r = (r + 1)(r + w)(r + w^2)$$

$$T_r = (r + 1)(r^2 - r + 1)$$

$$T_r = (r^3 + 1)$$

$$\Rightarrow \text{Sum} = \sum_{r=1}^{20} (r^3 + 1) = 44100 + 20 = 44120$$

12. Ans (B)

$$\begin{aligned} & |\vec{a} + \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} + \vec{a}|^2 \\ &= 2(\vec{a}^2 + \vec{b}^2 + \vec{c}^2 + \vec{a} \cdot \vec{b} - \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) \\ &= 2(9 + 16 + 25 + \vec{a} \cdot \vec{b} - \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) \\ &\vec{a} + \vec{b} + \vec{c} = \vec{0} \\ &\Rightarrow \vec{a} \cdot \vec{b} + \vec{c} \cdot \vec{a} = -\vec{a}^2 = -9 \\ &\text{Also } |\vec{b} + \vec{c}|^2 = 9 \\ &\Rightarrow 16 + 25 + 2\vec{b} \cdot \vec{c} = 9 \Rightarrow \vec{b} \cdot \vec{c} = -16 \\ &\therefore |\vec{a} + \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} + \vec{a}|^2 \\ &= 2(50 + 16 - 9) = 114 \end{aligned}$$

13. Ans (B)

Let direction ratios of line are a, b, c
Now $a + b + c = 0$
 $a - b + 2c = 0$
 $\frac{a}{3} = \frac{b}{-1} = \frac{c}{-2}$
So direction ratios of line are (3, -1, -2)
 $x + y + z + 1 = 0$ (A)
 $x - y + 2z + 1 = 0$ (B)
Put $z = 0$, we get $x = 1, y = 0$
Equation of line of intersection of $P_1 = 0, P_2 = 0$ is parallel to the plane $P_3 = 0$.
So distance of point (-1, 0, 0) from $P_3 = 0$ is
$$\left| \frac{-3 + 7}{\sqrt{9 + 1 + 16}} \right| = \frac{4}{\sqrt{26}}$$

15. Ans (D)

$p \equiv x^2 - 5x + 6 = 0$
 $q \equiv x = 3$
 $r \equiv x = -3$
Given statement is $p \rightarrow (q \vee r)$
It is equivalent to $\sim (q \vee r) \rightarrow \sim p$
 $\sim q \wedge \sim r \rightarrow \sim p$
 $\sim p$ and $\sim r \rightarrow \sim p$
i.e. option (D) is correct ans.

16. Ans (C)

Let us divide the outcomes into three separate cases :

I. Exactly one has even outcome

$$P_1 = \frac{{}^3C_1 \times 1 \times 3 \times 3}{6 \times 6 \times 6} = \frac{1}{8}$$

II. Exactly two have even outcomes

$$P_1 = \frac{{}^3C_1 \times 3 \times 3 \times 3}{6 \times 6 \times 6} = \frac{3}{8}$$

III. Exactly have even outcomes

$$P_3 = \frac{3 \times 3 \times 3}{6 \times 6 \times 6} = \frac{1}{8}$$

$$\therefore P = P_1 + P_2 + P_3 = \frac{5}{8}$$

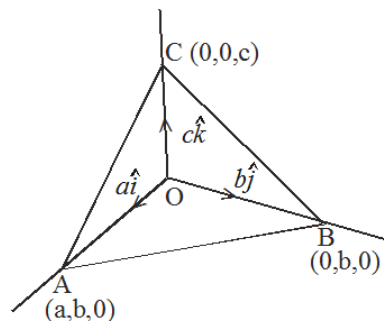
17. Ans (C)

$$P(E) = \frac{{}^8C_2 \cdot {}^8C_3 + {}^8C_3 C_2}{{}^9C_3 \cdot {}^8C_3}$$

18. Ans (B)

Let equation plane $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$
passing through (1, 1, 1)

$$\text{so, } \frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 1$$



Volume of tetrahedron OABC

$$\begin{aligned} &= \frac{1}{6} [a\hat{i} \ b\hat{j} \ c\hat{k}] \\ &= \frac{1}{6} abc \end{aligned}$$

because $a, b, c > 0$, so by G.M. $\geq$ H.M.

$$\sqrt[3]{abc} \geq \frac{3}{\frac{1}{a} + \frac{1}{b} + \frac{1}{c}}$$

$$abc \geq 27$$

$$\frac{1}{6} abc \geq \frac{9}{2}$$

$$V \geq \frac{9}{2}$$

19. Ans (B)

$$P(A \cap \bar{B} \cap C) + P(A \cap \bar{B} \cap \bar{C}) = P(A \cap \bar{B}) = 1/7.$$

$$P(A \cap B) = P(A) - P(A \cap \bar{B}) = 1/7.$$

$$P(\bar{A}/\bar{B}) = \frac{P(\bar{A} \cap \bar{B})}{P(\bar{B})} = \frac{1 - P(A \cap B)}{1 - P(B)} = \frac{4}{5}$$

20. Ans (C)

$$(z-1)(z-i)(z+2i) = 0$$

$$z_1 = 1, z_2 = i, z_3 = -2i$$

$$\operatorname{Re}(z_1) + \operatorname{Re}(z_2) + \operatorname{Re}(z_3) = 1$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (7.00)

There are four favourable combinations of three numbers

$$7+8+9, 6+8+9, 5+8+9, 6+7+9$$

$$\text{Hence, } n(E) = 4 \times (3 \times 3! \times 6!) \text{ \& } n(S) = 9!.$$

2. Ans (6.00)

$$\left| |z| - \left| \frac{5}{z} \right| \right| \leq \left| z - \frac{5}{z} \right| \Rightarrow \left| |z| - \frac{5}{|z|} \right| \leq 4$$

$$\text{Let } |z| = t$$

$$-4 \leq t - \frac{5}{t} \leq 4 \Rightarrow -4t \leq t^2 - 5 \leq 4t$$

$$\text{Now, } t^2 + 4t - 5 \geq 0 \quad t^2 - 4t - 5 \leq 0$$

$$(t+5)(t-1) \geq 0 \quad (t-5)(t+1) \leq 0$$

$$\Rightarrow t \in [1, \infty) \quad t \in [0, 5] \text{ (as } t \text{ is positive)}$$

$$\therefore t \in [1, 5]$$

3. Ans (8.00)

$$\vec{r} = a\hat{i} + b\hat{j} + c\hat{k}$$

$$\alpha = \frac{\vec{r} \cdot (2\hat{i} + \hat{k})}{\sqrt{5}} = \frac{2a + b}{\sqrt{5}}$$

$$\beta = \frac{-2a + b}{\sqrt{5}}, \quad r = c$$

$$\therefore |\alpha| = |\beta| = |\gamma|$$

$$\Rightarrow \underbrace{|2a + b|}_A = \underbrace{|-2a + b|}_B = \underbrace{|c\sqrt{5}|}_B$$

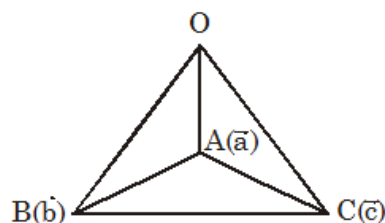
$$\left. \begin{array}{l} a = \pm \frac{\sqrt{5}}{3} \\ b = 0 \\ c = \pm \frac{2}{3} \end{array} \right\} \rightarrow 4$$

$$\left. \begin{array}{l} c = \pm \sqrt{\frac{1}{6}} \\ b = \pm \sqrt{\frac{5}{6}} \\ a = 0 \end{array} \right\} \rightarrow 4$$

$$a = 0$$

8 possible triplets.

4. Ans (2.00)



Opposite edges = OA, BC

$$|\vec{a}| = |\vec{b} - \vec{c}| = 6$$

$$\text{Shortest distance} = \frac{|\vec{a} \cdot (\vec{b} - \vec{c})|}{|\vec{a} \times (\vec{b} - \vec{c})|} = 4 \quad \dots(1)$$

$$|\vec{a} \times (\vec{b} - \vec{c})| = |\vec{a}| |\vec{b} - \vec{c}| \sin 30^\circ = 18^\circ$$

$$\text{Volume} = \frac{1}{6} [\vec{a} \cdot (\vec{b} \times \vec{c})] = 12 \text{ unit}^2$$

5. Ans (7.00)

Let equation of new plane

$$2x - 2y + z - 3 + \lambda z = 0$$

Point (3, 1, 1) lie on it $\Rightarrow \lambda = -2$

Hence equation of new plane $2x - 2y - z = 3$

$$\cos \alpha = \frac{4 + 4 - 1}{3.3} = \frac{7}{9}$$

6. Ans (3.00)

$$L_1 : \frac{x-2}{1} = \frac{y-9}{2} = \frac{z-13}{3} = \lambda$$

$$\Rightarrow x = 2 + \lambda, y = 2\lambda + 9, z = 3\lambda + 13$$

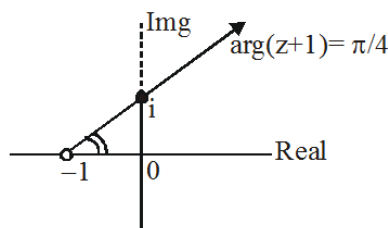
This lies on L_2

$$\therefore \underbrace{\frac{2+\lambda-a}{1}}_p = \underbrace{\frac{2\lambda+9-7}{-2}}_q = \underbrace{\frac{3\lambda+13+2}{3}}_r$$

by q and r : $\lambda = -3$

put in q and solve p and q, $a = -3$

7. Ans (1.00)



$$z^3 = \bar{z} \Rightarrow |z| = 0 \text{ or } 1$$

$$|z| = 0 \Rightarrow z = 0$$

$$|z| = 1 \Rightarrow z^4 = 1 \Rightarrow z = \pm 1, \pm i$$

only $z = i$ satisfies $\arg(z+1) = \frac{\pi}{4}$

8. Ans (11.00)

Position vector of diagonal $= 6\vec{a} - \vec{b}$

$$\text{Length of diagonal} = \sqrt{(6\vec{a} - \vec{b})^2}$$

$$= \sqrt{36|\vec{a}|^2 + |\vec{b}|^2 - 12(\vec{a} \cdot \vec{b})}$$

$$= \sqrt{36.8 + 9 - 12.2\sqrt{2}.3 \cdot \frac{1}{\sqrt{2}}}$$

$$= \sqrt{288 + 9 - 72}$$

$$= \sqrt{297 - 72} = \sqrt{225}$$

Position vector of other diagonal $= 4\vec{a} + 5\vec{b}$

$$\text{Length of diagonal} = \sqrt{|4\vec{a} + 5\vec{b}|^2}$$

$$= \sqrt{16|\vec{a}|^2 + 25|\vec{b}|^2 + 2.4|\vec{a}| \cdot 5|\vec{b}| \cdot \frac{1}{\sqrt{2}}}$$

$$= \sqrt{16.8 + 25.9 + 40.32}$$

$$= \sqrt{128 + 225 + 240} = \sqrt{593}$$

$$\alpha = 5, \beta = 9, r = 3$$

$$\alpha + \beta + r = 5 + 9 - 3 = 11$$

9. Ans (4.00)

$$|z_1 + z_2| = |z_1| + |z_2| \Rightarrow z_1, z_2, 0 \text{ are collinear}$$

$$z_1 = rz_2 (r > 0)$$

$$\frac{a^2}{b} = \frac{(z_1 + z_2)^2}{z_1 z_2} = \frac{(1+r)^2}{r} = \frac{1}{r} + r + 2 \geq 4$$

10. Ans (3.50)

Let A be origin $\overrightarrow{AB} = \vec{a}$, $\overrightarrow{AD} = \vec{b}$

$$\text{so, } \overrightarrow{AE} = \vec{b} + \frac{3}{2}\vec{a}, \overrightarrow{AG} = \vec{a} + 3\vec{b}$$

$\Rightarrow$ the required ratio

$$= \frac{\frac{1}{2} |(\vec{a} + 3\vec{b}) \times (\vec{b} + \frac{3}{2}\vec{a})|}{\frac{1}{2} |\vec{a} \times \vec{b}|} = \frac{7}{2}$$